* 0000800000001 *

Cambridge Assessment International Education

Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

8250696578

BIOLOGY **5090/41**

Paper 4 Alternative to Practical **October/November 2024**

1 hour

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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1 Yeast breaks down sugar (glucose) to provide energy for growth and reproduction by respiring anaerobically:

glucose ⟶ carbon dioxide + alcohol

Sugar is used in human food to make it taste sweet. Too much sugar in the diet can cause diseases. Sugar substitutes are available which taste as sweet as sugar.

A student decided to investigate whether yeast can use a sugar substitute for respiration.

Three test-tubes were set up as shown in Table 1.1.

Table 1.1

contents	**mixture in the test-tubes**		
distilled water / cm^3	15	15	15
yeast / g	1	1	1
test substance	1 g sugar	1 g sugar substitute	no addition

The three test-tubes were labelled so that they could be identified.

(a) (i) State the labels that you would use on the three test-tubes.

test-tube containing sugar: .....

test-tube containing sugar substitute: .....

test-tube with no addition: .....

[1]

A stirring rod was used to mix the contents of each test-tube.

(ii) Explain why it was important to clean the stirring rod after using it in each test-tube.

.....

..... [1]

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The level of the mixtures in each test-tube was marked with a water-resistant marker. This was the starting level.

When the yeast respires, it produces bubbles of gas that are trapped in the mixture, making the level of the mixture rise up the test-tube.

The student used a large beaker as a water-bath to control the temperature of the test-tubes in this investigation. The temperature of the water at the beginning was 35 °C. The aim was to maintain this temperature throughout the investigation by adding hot or cold water.

The three test-tubes were placed in the water-bath, and a timer was started. At 5 minutes, 10 minutes and 15 minutes, the distances moved by the mixtures from the starting levels were measured and recorded. The temperature of the water-bath was also recorded.

Fig. 1.1 shows the lower parts of the test-tubes at 5 minutes and the water-bath thermometer reading at 5 minutes.

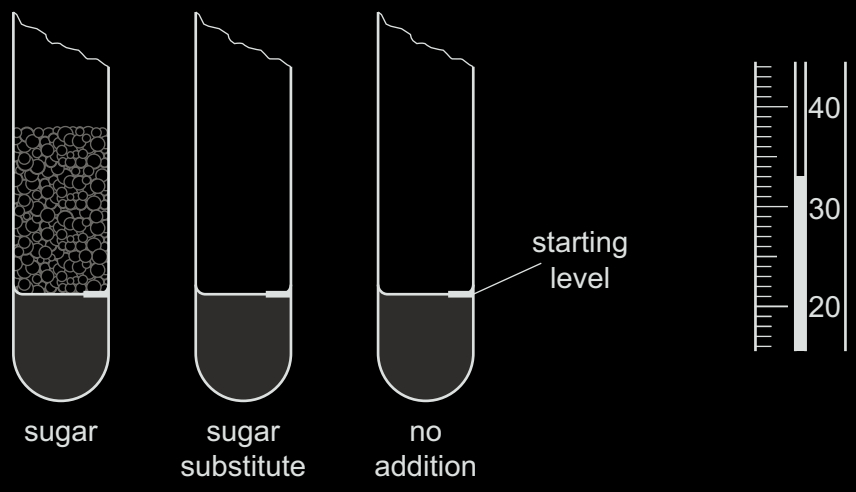


Fig. 1.1

Fig. 1.2 shows the results for 10 and 15 minutes recorded in the student's notebook.

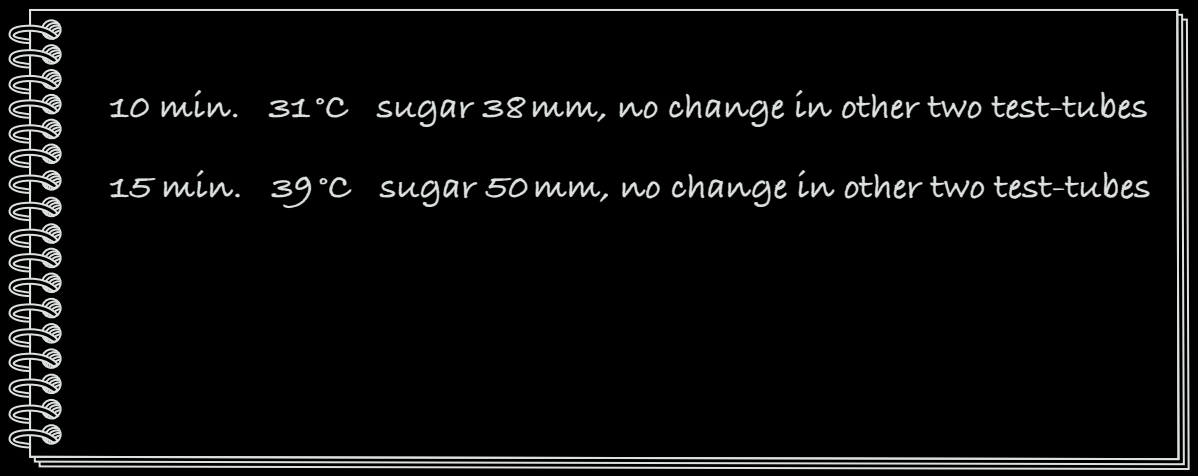


Fig. 1.2

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(b) (i) Complete Table 1.2:

- Complete the headings.
- Using Fig. 1.1, measure the distances moved by the yeast mixtures from the starting levels in the three test-tubes and record them. Read the temperature shown on the water-bath thermometer in Fig. 1.1 and record it.
- Use the information in Fig. 1.2 to complete Table 1.2.

Table 1.2

.....	.....			temperature of water-bath / °C
	sugar	sugar substitute	no addition	
5				
10				
15				

[5]

(ii) Explain how the test-tube with no addition acted as a control in this investigation.

.....

..... [1]

(iii) State what conclusion you can make from the results of this investigation.

.....

.....

.....

..... [1]

(iv) Use the information in Fig. 1.1 and Fig. 1.2 to evaluate the student's control of the water-bath temperature.

.....

..... [1]

(v) Suggest why the temperature of the water-bath at 15 min was 39 °C.

.....

..... [1]

(c) The yeast's activity could also have been measured by recording any increase in the volume of the mixtures in the test-tubes.

Fig. 1.3 shows a test-tube with sugar, yeast and distilled water as seen from above. The distance above the marked starting level in this test-tube after 10 minutes was 3.8 cm.

The line between **A** and **B** indicates the diameter of the mixture within the test-tube.

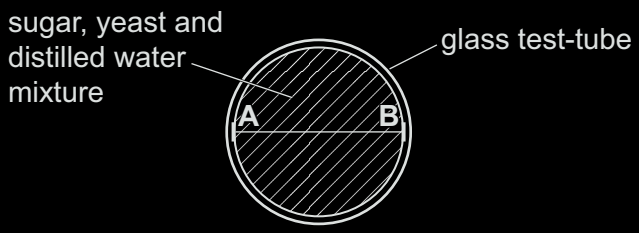


Fig. 1.3

(i) Measure and record the length of the line between **A** and **B**.

diameter of mixture within test-tube ..... cm

Calculate the increase in volume of the mixture in this test-tube after 10 minutes.

Use 3.14 as the value of π. Give your answer to 1 decimal place.

Show your working.

volume increase ..... cm^3
[4]

(ii) The apparatus used in this investigation is shown in Fig. 1.1. Suggest a piece of apparatus that could have been used in this investigation to directly measure the volume of the mixture. Describe how you would determine the increase in volume at each time interval using this apparatus.

apparatus .....

determination of increase in volume .....

.....

.....
[2]

[Total: 17]

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2 When organisms respire aerobically they use oxygen and produce carbon dioxide.

Some students investigated the rate at which germinating seeds respired, using the apparatus in Fig. 2.1.

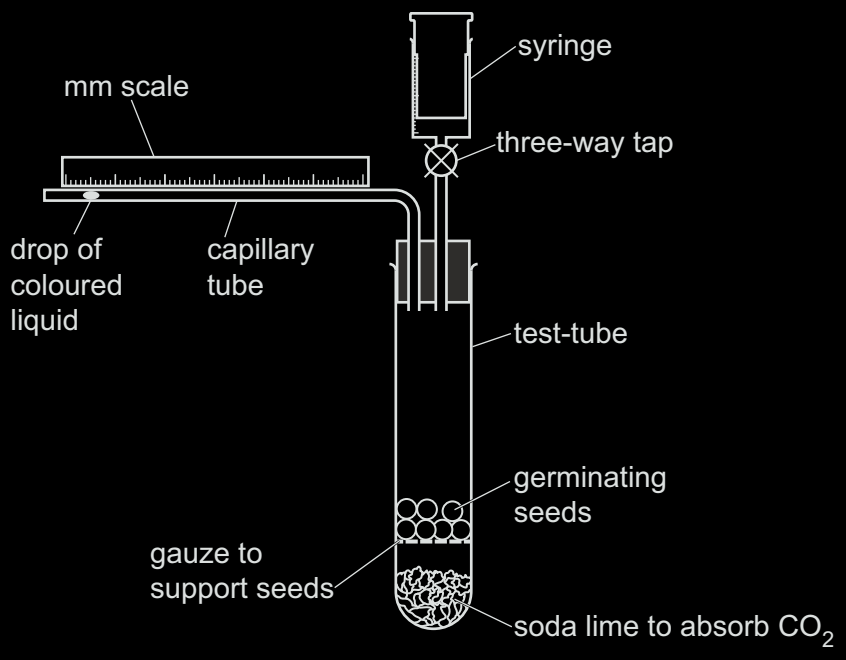


Fig. 2.1

Carbon dioxide produced by the germinating seeds was absorbed by the soda lime. As oxygen was used the volume of gas in the apparatus reduced and the drop of coloured liquid moved along the capillary tube towards the seeds.

The students moved the drop of coloured liquid in the capillary tube to the beginning of the scale (0 mm) by opening the three-way tap and using the syringe to carefully push air into the apparatus. They then closed the tap.

This was the start position for the drop of coloured liquid. Its position on the scale was recorded over the next four minutes. The movement of the drop of coloured liquid indicates the rate of respiration of the seeds.

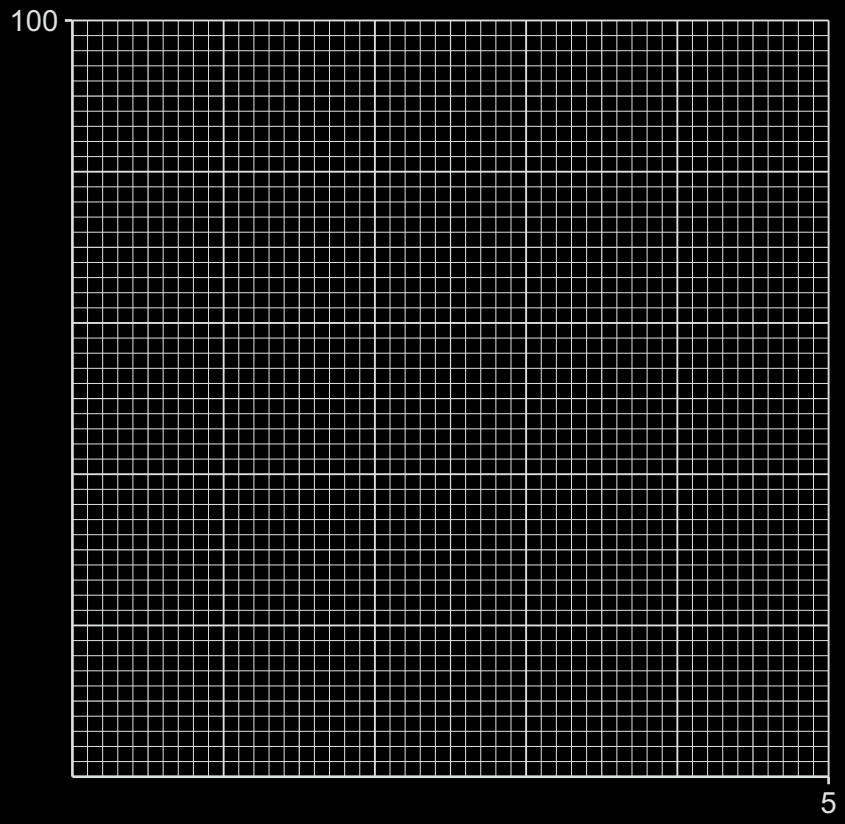
The students' results are shown in Table 2.1.

Table 2.1

time / minutes	position of drop of coloured liquid / mm
0	0
1	18
2	36
3	54
4	72

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(a) (i) Construct a line graph of the data in Table 2.1 on the grid. The values for the end points of the axes are shown on the grid. Draw a straight line of best-fit to connect the points.



[4]

(ii) Use your graph to predict the position of the drop of coloured liquid at 5 minutes. Show your working on the graph.

position ..... mm [2]

(iii) Use the result at 4 minutes in Table 2.1 to calculate the rate of movement of the drop of coloured liquid caused by the respiration of the seeds.

Space for working.

rate of movement ..... [2]

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(b) Plan an investigation to determine the effect of temperature on the rate of respiration in germinating seeds. Use the apparatus in Fig. 2.1 in your plan.

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..... [6]

[Total: 14]

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3 Fig. 3.1 shows the whole of a one-seeded fruit of a dandelion flower. A single dandelion flowerhead can produce up to 200 of these one-seeded fruits. The mass of a single fruit is 0.0005 g.

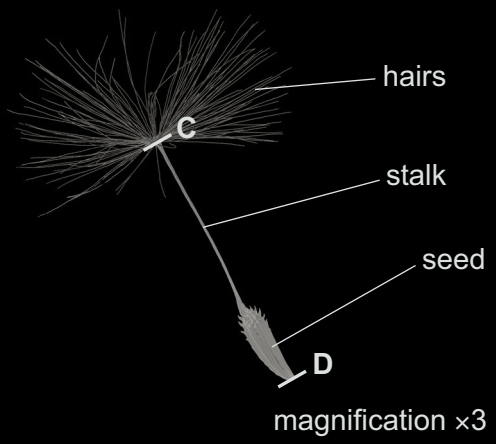


Fig. 3.1

(a) (i) In the space below make a large drawing of the whole fruit as it appears in Fig. 3.1.

[4]

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(ii) The lines **C** and **D** indicate the total length of the seed and stalk. Draw a straight line on Fig. 3.1 to join **C** and **D**. Measure the length of the line and record it.

.....

Calculate the **actual** length of the seed and stalk and record it to the nearest whole number.

Space for working.

..... mm [3]

(b) Identify **two** features of this fruit that show it is adapted for dispersal by wind. Explain your answers.

1 .....

.....

2 .....

.....

[2]

[Total: 9]

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